



SIMATS
ENGINEERING



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Sreeetha Institute of Medical And Technical Sciences
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TASK SCHEDULING

Task Scheduling System Using Priority Queue

Selecting an Efficient Data Structure for Job Scheduling

CSA0302- DATA STRUCTURES

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Introduction

What is a task scheduling system?

- 1 A task scheduling system manages multiple jobs.
- 2 Each job has a priority level.
- 3 High-priority jobs should execute before low-priority jobs.
- 4 The choice of data structure affects system performance.

EXAMPLE

P1

Priority 1

Emergency Task

P2

Priority 2

System Task

P3

Priority 3

Background Task

Problem Statement

OBJECTIVE

- Execute jobs based on priority.
- Support quick insertion of new jobs.
- Retrieve the highest-priority job efficiently.
- Handle a large number of tasks with minimum delay.

CHALLENGES



Fast Scheduling

Jobs must be picked up with minimal delay.



Efficient Memory Usage

Structure should not waste resources.



Scalability

Must handle growth in job volume smoothly.

Suitable Data Structure

Priority Queue (Binary Heap)

WHY?

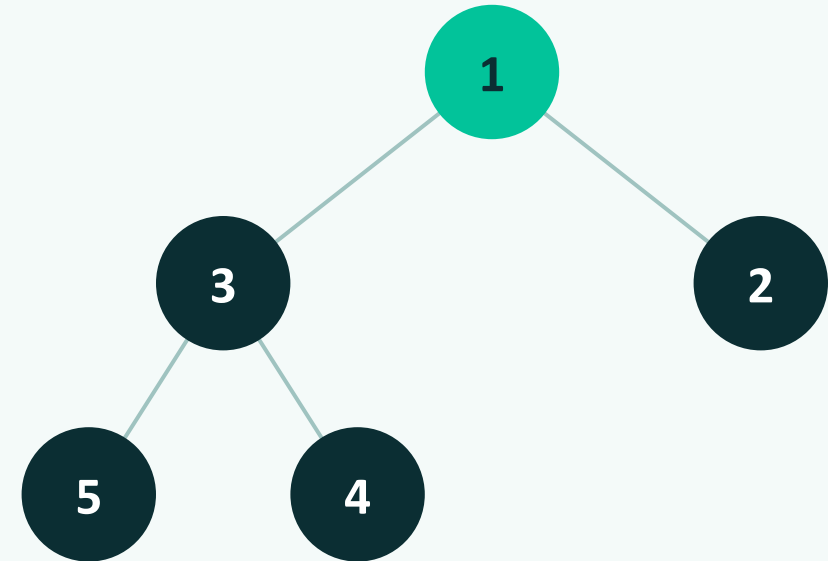
- ✓ Stores jobs according to priority.
- ✓ Highest-priority job is always at the top.
- ✓ Faster than sorting the entire list every time.

OPERATIONS

Insert Job

Remove Highest Priority

Peek Highest Priority



Min-Heap — smaller number = higher priority

Performance Analysis

Operation	Time Complexity
Insert	$O(\log n)$
Delete Highest Priority	$O(\log n)$
Peek Highest Priority	$O(1)$
Build Heap	$O(n)$

ADVANTAGES

- ✓ Fast execution
- ✓ Efficient scheduling
- ✓ Suitable for real-time systems
- ✓ Handles dynamic job arrivals

REAL-TIME SUITABILITY



Operating Systems



Cloud Computing



Network Scheduling



Printer Queues

A heap keeps scheduling overhead low even as job volume grows — ideal where delay directly affects users.

Working Example

JOBS

Job	Priority
A	3
B	1
C	2
D	5
E	4

Smaller number = Higher Priority

EXECUTION ORDER

1

Job B

Priority 1

2

Job C

Priority 2

3

Job A

Priority 3

4

Job E

Priority 4

5

Job D

Priority 5

Conclusion

- ✓ Priority Queue (Binary Heap) is the most suitable data structure for task scheduling.
- ✓ Provides efficient insertion and deletion.
- ✓ Reduces execution delay.

WIDELY USED IN

Operating Systems

CPU Scheduling

Printer Scheduling

Network Packet Scheduling

Cloud Computing

Priority Queue ensures efficient, scalable, and high-performance task scheduling with $O(\log n)$ insertion and deletion — making it the best choice for execution based on priority.